



PLANETARY AUTONOMOUS LANDING ZONE ASSESSMENT

& HABITAT PLANNING SYSTEM

Physics-informed survey, hazard scoring, illumination analysis, communications analysis, and habitat planning for future planetary operations.



WHAT IS THIS APPLICATION?

An integrated mission planning tool that analyzes planetary terrain, assesses hazards, evaluates illumination and communications, plans safe rover routes, and recommends habitat locations. Outputs are suitable for mission concept development and systems engineering trade studies.



AUTONOMOUS
SURVEY



TERRAIN
GENERATION



ILLUMINATION
ANALYSIS



COMMUNICATIONS
ANALYSIS



HABITAT
PLANNING



MISSION
OPTIMIZATION

HOW IT WORKS: 4-STEP MISSION WORKFLOW

1 CONFIGURE



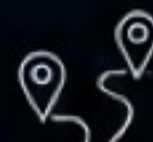
Select target body,
set mission parameters,
and analysis options.

2 ANALYZE



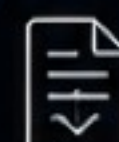
Generate terrain and run
physics-based analyses for
hazards, illumination, and
communications.

3 PLAN



Identify best landing zone,
plan rover survey routes,
and recommend habitat
and infrastructure.

4 EXPORT



Export mission reports,
maps, and data for
downstream engineering
and operations.

KEY CAPABILITIES



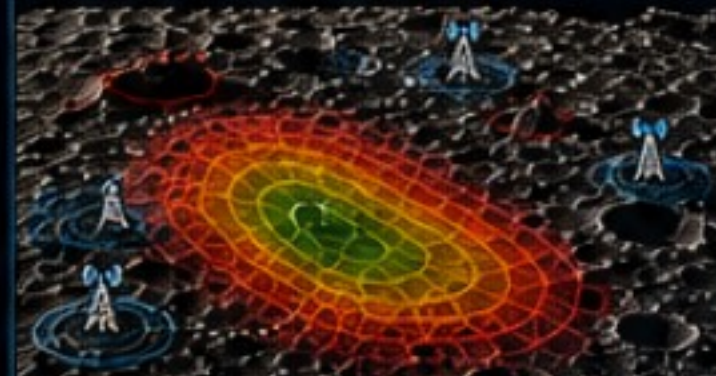
PHYSICS-BASED HAZARD ASSESSMENT



- Detects & scores craters, boulders, slope hazards, and terrain shadows
- Combines slope, hazards, illumination, and communications into a single Landing Suitability Score (0-100)
- Recommendations for optimal landing zones



ILLUMINATION ANALYSIS



- Sun vector geometry and incidence angle calculation
- Shadow fraction estimation
- Illumination index (0-1) for solar power and visibility assessment



COMMUNICATIONS ANALYSIS



- Line-of-sight to relay network with terrain masking
- Relay azimuth/elevation & blocked fraction
- Communications quality index (0-1) with deep-space penalty



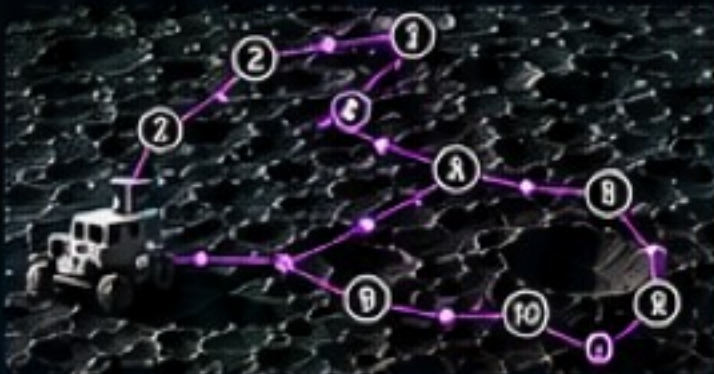
THERMAL ASSESSMENT



- Surface environment thermal model
- Thermal penalty and thermal environment index (0-1)
- Day/night and body-specific thermal considerations



HAZARD-AWARE ROVER PATH PLANNING



- A* path planning with hazard cost map
- Waypoints, path length, estimated time, and coverage
- Safer routes that avoid high-risk areas



HABITAT & INFRASTRUCTURE PLAN



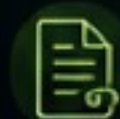
- Recommends Habitat Zone, Power Zone, and Comms Relay
- Optimized locations based on terrain, illumination, and communications
- Exportable coordinates and maps



MULTI-BODY SUPPORT



- Moon • Mars • Phobos
- Europa • Custom Airless Body
- Body-specific gravity, solar flux, and constants
- Easy switching between targets



MISSION EXPORT & REPORTING



- Mission Report (JSON)
- Mission Summary (CSV)
- Terrain & Hazard Maps (PNG)
- Recommended Coordinates (CSV)
- Ready for engineering workflows

OPTIONAL GPT-5.5 ADVISORY LAYER (ADVISORY ONLY)



AI Advisory
Provides mission-specific
insights and recommendations



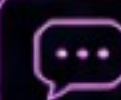
Physics-First
All recommendations are based on
physics-informed model outputs



Does Not Modify Results
AI never overrides or changes
physics calculations



Human in the Loop
Human verification and validation
remain required



EXAMPLE AI INSIGHT

"The candidate zone on the eastern rim offers high illumination and strong communications, but contains boulder clusters. The central depression zone provides smoother terrain with lower slopes and better thermal stability, making it the safer landing option."

SUPPORTED PLANETARY BODIES



MOON



MARS



PHOBOS



EUROPA



CUSTOM
AIRLESS BODY



IMPORTANT DISCLAIMER

This application is a physics-informed conceptual assessment tool intended for mission concept development and systems engineering studies. It is not a substitute for high-fidelity engineering analysis, detailed geotechnical surveys, or mission verification. Not flight-certified analysis.

